



B.Tech : I & II Semester

Programming for problem Solving

:~: Lab & Practice Exercises :~:

[Simple Calculations/Formula Based/using different operators]

Assignment : I

1. Write a program to add any two integer values and display sum of the given numbers.
2. Write a program to enter three students age, find out Average age of the students.
3. Write a program to enter three variables of different data types, and display in the own data type format.
4. Write a program to compute the volume $\frac{4}{3} \pi r^3$ and surface area ($4\pi r^2$) of a sphere of any radius r.
5. Write a program to compute the perimeter, P and Area A of a triangle of sides a, b and c. where
 $p = a+b+c$, $A = \sqrt{s(s-a)(s-b)(s-c)}$, $2s = a + b + c$.
6. Write a program to compute the roots of the quadratic equation $ax^2 + bx + c = 0$. $X = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
7. Write a program to input the temperature in Degree Fahrenheit F, Convert them into Degree Celsius C.

$$C = \frac{5(F - 32)}{9}$$

8. Write a program to compute the area of the triangle when the Base and Height of the triangle are given. [Hint : Area = $\frac{1}{2} * \text{Base} * \text{Height}$]
9. Write a program to enter roll no., name of the student, and five subject's marks then compute the total obtained marks, percentage. Display all the things also.
10. Write a program to display your First Name, Middle Name, and Surname. Now display your complete name in same line, and also display different lines.
11. Write a program to compute the Net Salary of an employee. Enter the employee name and Basic Salary of an employee, now compute the all allowances.

HRA	:	15% of Basic Salary	SI	:	200
DA	:	23% of Basic Salary	Total Deduction	:	PF + SI
PF	:	10% of Basic Salary	Net Salary = Gross Salary - Total Deduction.		
Gross Salary = Basic Salary + HRA + DA					
12. Write a program to enter the circumference of a circle then find out the Area of the given circle.
13. Write a program to read an integer value of four digits. Now find out the sum of the value for each digit.
[Hint : n=1234 , Sum = 10]
14. Write a program to read an integer value of four digits. Now display the entered number in reverse order.
[Hint : n=1234 → 4321]

15. Write program to find out the largest and smallest number of 2 and 3 numbers, using Conditional Operator (?).
16. Write a program to swap the values in two integers. [Hint : a= 10, b=20 : a=20, b=10]
17. Write a program to printing the ASCII value of char input.
18. Write a program to enter the number of days. Now find out the how many Years [Y], Months [M]and Weak [W].
19. Write a program that accepts input of number of seconds, validates it and output the equivalent number of hours, minutes and seconds.
20. Write a program to input Principal amount P and the Annual Rate of interest r. Compute the interest I.

$$i = p \left(1 + \frac{r}{100} \right)^n - p \quad \text{Where n is the year.}$$

[Using IF..... /IF.....Else/ Nested..... IF/ LadderIF Based]

Assignment: 2

1. Write a program to enter values [int/float] and find out Largest and Smallest elements.
 1. For two input values.
 2. For three input values.
 3. For four input values.
2. Repeat the exercise no. : 1, using conditional or Ternary Operator?
3. Write a program to enter a number, check the entered number is even or odd display the number with appropriate message.
4. Write a program to enter a numeric value then check the entered number is Positive, Negative or Zero.
5. Write a program to enter a number, check the entered number is even or odd. Now the compute square of the given number, if number is even, or compute the cube when number is odd. [Hint: Even 2= 4, Odd 3=27]
6. Write a program to enter an integer's value and display the message the number is divisible by 7 or not.
7. Write a program to three input values find out the largest element also check the condition when a=b=c.
8. Write a program to check the whether a year is LEAP year or not.
9. Write a program to calculate gross salary of the person. Given Basic Salary (bs) as input. If bs > 5000, da = 55% of bs and hra = 15% of bs else da = 45% of bs and hra = 10% of bs.
11. Write a program to compute an Electricity Bill, Enter Consumer No., Consumer Name, and Consumed Unit.

CONSUMED UNIT	CHARGES
➤ 0 – 200	Rs. 0.50/- per unit
➤ 201 – 400	Rs. 0.65/- per additional unit
➤ 401 – 600	Rs. 0.80/- per additional unit
➤ 600 and above	Rs. 1.00/- per additional unit.

Display the Consumer Number, Consumed Unit and Payable Amount [To be pay].

12. Write a program to compute the grade of student to the following rules.

AVERAGE MARKS

GRADE

- | | |
|-------------|-----------------|
| ➤ 80 to 100 | Honors |
| ➤ 60 to 79 | First Division |
| ➤ 50 to 59 | Second Division |
| ➤ 40 to 49 | Third Division |
| ➤ 0 to 39 | Fail |

Enter the roll no, Name and three subjects' marks compute the total, percentage of the obtained marks and display all the things.

Write a program to determine the status of entered character. It is Upper case letter, Lower case letter, digit or nay special character. [ASCII Code : Upper case 65-90, Lower case 97-122, Digit 48-57, Special Character 0-47, 58-64,91-96,123-124].

13. Write a program to enter any letter and display with its change case [case conversion]. [HINT : ASCII code 65 to 90 UPPER CASE, 97 to 122 LOWER CASE]
14. Write a program to enter any number and display message, the entered number is belong to a particular range. The ranges are divided into four ranges (0-25), (26-50), (51-75) and (76-100).
15. Write a program to check for any candidate is eligible or not for admission in University, the admission criteria are as follows: the Obtained Marks in Mathematics >= 60, Obtained Marks in Physics >= 50, Obtained Marks in Chemistry >=40 and total aggregate marks in all three subjects must be >=200,

otherwise Total Obtained Marks in Physics and Chemistry ≥ 150 . Enter three subjects' marks and display message "Eligible for admission" or "Not Eligible for admission".

16. Write a program to enter three sides of a triangle, and check given triangle is "Valid Triangle" or "Not Valid Triangle". The triangle is valid if sum of the two sides is greater than the largest side of the triangle.
17. Write a program to enter three angles of a triangle. Now check given triangle is and check given triangle is "Valid Triangle" or "Not Valid Triangle". A triangle is valid if the sum of all three angles is equal to 180.

[Using SWITCH CASE Statement]

1. Write a program to compute the simple mathematical operations like Addition/Subtraction/Multiplication & Division of any two numbers.
2. Write a program to enter the no of day and display its name of the day respectively.
[Hint : Monday =1, Tuesday =2, Wednesday =3, Thursday =4, Friday = 5, Saturday =6, Sunday =7]
3. Write a program to conversions of the temperature. Display the output screen with different choices :
[1] For Centigrade to Fahrenheit
[2] For Fahrenheit to Centigrade
[3] Exit
Enter Your Choice.....:
4. Write a program to check given character is Vowel or Consonant using switch statement.
5. Write a program to enter a Digit [0 to 9] and print it in words. [Hint : 2= TWO]
6. Write a program to enter a First Character of the Color, and then display the respective color.
[Hint : 'R' or 'r' =RED]
7. Write a program to compute the Addition, Subtraction, Multiplication and Division of two numbers with entering the **appropriate sign**. [Hint : + for addition, - for subtraction, * for Multiplication and / for Division].
8. Write a menu driven program to compute area of different shapes, like Circle, Triangle, Rectangle, and Square etc. using the switch case statements.

```

===== MAIN MENU =====
1.      :      Area of Circle
2.      :      Area of Triangle
3.      :      Area of Rectangle
4.      :      Area of Square
5.      :      Exit
=====
:-:      Enter your choice .....[1 to 5]:
```

[Simple Loop based program i.e. While ()/ Do.....while ()/For (; ;)

Assignment : 3]

Compute Sum of the given series:

1. $1+2+3+4+5+6+\dots+n$ Sum = ?
2. $1/1+1/2+1/3+1/4+\dots+n$ Sum = ?
- 3.

1. Write a program to generate and compute Sum of the series terms. All different series are given below.
 4. 1, 2, 3, 4.....n Sum = ?
 5. 50, 51,100 Sum = ?
 6. 100, 99,1 Sum = ?
 7. 2, 4, 6, 8.....n Sum = ?
 8. 1, 3, 5,n Sum = ?
 9. 1, 4, 9,n Sum = ?
 10. 1, 8, 27,n Sum = ?
2. Write a program to draw a multiplication table of any integer N, the multipliers are 1, 2, 3.....10.
3. Write a program to generate the **Fibonacci Series** up to n terms. [1, 1, 2, 3, 5, 8, 13,n].
4. Write a program to find the Factorial of any number n.
5. Write a program to Input the positive integer number
 11. Display the number in reverse order [1234 = 4321]
 12. Compute sum of the each digits. [1+2+3+4 = 10]
 13. Compute multiplication value of each digit. [1*2*3*4 = 24]
6. There are m groups, each group having n numbers. Write a program to find out the average of each group separately. Input the one by one.
7. Write a program to 2 D multiplication Table for 10 numbers from 1 to 10 in step of 1. The multipliers are 1, 2, 3....12. **[Using Nested Loop Statements]**
8. Write a program to count the Odd and Even numbers, between 1 to 100. Find out sum of the all even & odd numbers separately.
9. Write a program to find out the divisors of a given numbers. [Hint: 18 : 1, 2, 3, 6, 9]
10. Write a program to find the numbers and sum of the all integers greater than 100 and less than 200 that are divisible by 7.
11. Write a program to enter any integer value now check the entered number is ARMSTRONG NUMBER or NOT.
12. Write a program to generate a series of Armstrong numbers. A number is an Armstrong Number, if sum of the cube of its digit is equal to the number.
13. Write a program to find reverse of a numbers and check whether it is palindrome or not.
14. Write a program to compute $y = 2x^4 + 5x^3 + 6x^2 - 8x + 9$, For x= -10,-9,-8.....8,0,10
15. Input Principal Amount p, and the annual rate of interest r, Write a program to compute the interest I for 1 to 10 years in step of 1 $i = p(1 + \frac{r}{100})^n - p$
16. There are m group each group having n numbers. Write a program to find th average of each group separately . Input the number one by one.
17. Write program to print the following patterns / Out puts. **[Using Nested Loop Statements]**

a.

```

1
2 2
3 3 3
4 4 4 4
    
```

b.

```

1
1 2
1 2 3
1 2 3 4
    
```

c.

```

1
1 2
1 2 3
1 2 3 4
    
```

d.

```

1
2 2
3 3 3
4 4 4 4
    
```

e.

```

A
B C
D E F
G H I J
    
```

f.

```

A
A B
A B C
A B C D
    
```

g.

```

1
2 3
4 5 6
7 8 9 10
    
```

h.

```

1
2 2
3 3 3
4 4 4 4
    
```

i.

```

1
2 2 2
3 3 3 3
4 4 4 4 4 4
    
```

j.

```

1
0 1
1 0 1
0 1 0 1
    
```

k.

```

0
1 0
0 1 0
1 0 1 0
    
```

l.

```

*
* *
* * *
* * * *
    
```

1. Write a program to read an array of n elements, now display entered elements in array :
 - Same or Entered Order.
 - Reverse Order.
 - Compute sum of the all elements.
2. Write a program to read an array of n elements, Compute the **Average** of the entered elements.
3. Write a program to read an array of n elements. Find out the Largest and Smallest elements with Location in the array.
4. Write a program to read an array of n elements. Find out a particular element and replace with any new element (999).
5. Write a program to read an array of the n elements. Now display the occurrence of any particular element in array.
6. Write a program to read an array of n elements. Arrange all the elements. **[Using Bubble/Exchange Sorting Method].**
 14. [Ascending Order.]
 15. [Descending Order.]
7. Write a program to read an array of n elements. Find out a particular element in an array with his location.
 16. [Linear Search].
 17. [Binary Search].
8. Write a program to read two different array a[] and b [], now perform the merge operation and add all elements in new array in c[]. [a [] M elements, b [].....N elements: Result .. c [] M+N]

[Arrays : Two Dimensional]

9. Write a program to a matrix of [m x n] order, display and perform the following tasks:
10. Find out the sum of diagonal elements.
11. Square of the diagonal elements.
12. Display the Upper and Lower Half elements of the given matrices.
13. Scand Array :

1	2	3
4	5	6
7	8	9

Desired Array:

1	2	3	1	0	0	0	1	1
4	25	6	0	5	0	1	0	1
7	8	81	0	0	9	1	1	0
14. Write a program to read and display two matrices and compute the Addition & Subtraction of the given matrices.
15. Write a program to read and display two matrices, compute the Multiplication of the given matrices.
16. Write a program to a matrix of [m x n] order, compute the Transpose the given matrices.
17. Write a program to a matrix of [m x n] order, and perform the following tasks :
 - Compute Sum of the each Row's Element.
 - Compute Sum of the each Column's Element.

18. Write a program to a matrix of [m x n] order. Find out the Largest and Smallest element in the 2D array; also find out the location of an element in array.

[STRING]

ASSIGNMENT: 05

1. Write a program to read a text of line, display the entered text in same and reverse order.
[By using string function or without function]
2. Write a program to read a text of line, display in the entered form, now change the case of text.
[lower case to upper case or vice versa] [Hint : entered text :: JnU JaiPUr → out put :: jNu jAlpuR]
3. Write a program in 'C' to read and display strings, now check the purpose of the given functions :
 - i. strcat()
 - ii. strcmp()
 - iii. strcmpi()
 - iv. strcpy()
 - v. strlen()
 - vi. strrev()
 - vii. strlwr()
 - viii.strupr()
 - ix. strchr()
 - x. strrchr()
 - xi. strstr()
 - xii. strset()
 - xiii. strncat()
 - xiv. strncmp()
 - xv. strncpy()
 - xvi. strnset()
4. Write a p program to read and display a string, now check entered string is Palindrome or not.
5. Write a program to read and display a paragraph (multiple lines of text), now count and display following things :
 - i. Total number of characters.
 - ii. Total number of lines.
 - iii. Total number of words.
 - iv. Total number of blank spaces.
6. Write a program to read and display a paragraph (multiple lines of text), now count and display following things. :
 - i. Total number of Vowels.
 - ii. Total number of Consonants
 - iii. Total number of digits.
 - iv. Total number of Special Characters.
7. Write a program to enter the name of the students, display in before and after the sorting.
 - i. Ascending Order
 - ii. Descending Order
8. Write a program to read a text of line, display the entered text, now find out occurrence of any particular character in the given text.
9. Write a program to enter and display list of student's name, now find any particular name and replace it with new one. Display the modified list of the students.

10. Write a program to read a list of names, display the entered name, now find out occurrence of any particular name in given list.

[FUNCTION (USER DEFINED FUNCTIONS)]

ASSIGNMENT : 06

1. First of all, try to convert all maximum number of programs by defining user defined functions. [which you have performed in Assignment : 1, Assignment : 2, Assignment : 3, Assignment : 4].
2. Write a program by defining the user defined function, to enter three different values, Compute the **Average** of the entered elements.
3. Write a program by defining the user defined function, to compute factorial value of nay positive integer value.
4. Write a program by defining the user defined function, to generate a Fibonacci Series up to Nth element.
5. Write a program to read an array of n elements, Compute the **Average** of the entered elements.
6. Write a program to read an array of n elements. Find out the Largest and Smallest elements in array.
7. Write a program to read an array of n elements. Arrange all the elements.

[Using Bubble/Exchange Sorting Method].

- [Ascending Order.]
 - [Descending Order.]
8. Write a program to read an array of n elements. Find out a particular element in an array with his location.
 - [Linear Search].
 - [Binary Search].
 9. Write a program to by defining the user defined function, to swap the values in two integers, by using method of : [Hint : $a=10, b=20 : a=20, b=10$]
 - a. Call by Value.
 - b. Call by Reference.

10.